

vii. The Mars

*Mars is the fourth planet from the sun third being Earth.

*The axial tilt of Mars is $25^{\circ}12'$ relative to its orbital plane.

*Mars average distance from the sun is roughly 227943824 km (228 million km) and its orbital period is 687.0 Earth days. The diameter of Mars is 6792 km and it takes 24.7 hours to complete one rotation. *The atmospheric composition of Mars is CO_2 - 95.1%, N_2 -2.59%, Argon - 1.94%, Oxygen - 0.16% and traces of water. *NASA's **Mars Reconnaissance Orbiter** (MRO), took pictures of Mars with the help of its hi-tech cameras. After studying the pictures, scientists were able to claim the presence of liquid water on Mars. *The MRO contains a host of scientific instruments such as imaging spectrometers that helped in finding the hydrated minerals of the planet. The average temperature of Mars, it is -65°C (208 K). *Mars and Earth are very different planets when it comes to temperature, size and atmosphere, but original processes on the two Planets are surprisingly similar. *Mars is often referred to as the **Red planet** because the reddish iron oxide prevalent on its surface gives it a reddish appearance. *Mars has two relatively small natural moons, **Phobos** and **Deimos**. *The Nix Olympia mountain (volcano) of Mars is approx. three times the size of our Mount Everest. *The **Phoenix Mars Lander** was a space exploration mission of NASA on Mars under the Scout program. It was launched on August 4th, 2007 and landed on Mars on May 25th, 2008. The mission was declared concluded on November 2, 2008. Phoenix was the first lander that gave valuable inputs regarding the **presence of ice** on the planet. *The **Curiosity Rover** was launched by NASA to explore the possibility of life, determine the role of water, and study the climate and geology of Mars. The Mars Orbiter Mission (MoM) or **Mangalyaan** is India's first Mars exploration mission. It was launched on 5th November 2013 by the Indian Space Research Organization (ISRO). It was launched with the help of the PSLV- C25 rocket. It was inserted into Mars's orbit on 24 September 2014. Now, with the success of the MoM mission India has become a member of the elite Mars club (United States of America, Russia, European Union).

1. The duration of its day and tilt of its axis are almost identical to those of the Earth. This is true of :

- (a) Uranus (b) Neptune
(c) Saturn (d) Mars

U.P.P.C.S.(Pre) 2001

I.A.S. (Pre) 1994

Ans. (d)

The duration of the day on Earth is 23 hours, 56 minutes 4 seconds and the tilt of its axis is $23^{\circ}30'$. The planet Mars has nearly the same value as Earth for its day duration which is 24 hours, 37 minutes 25 seconds and Mars tilt of the axis is $25^{\circ}12'$. The duration of the day and tilt of the axis of other planets is as follows-

Uranus – duration of its day – 17 hours 14 minutes
the tilt of its axis – 97.77°

Neptune – duration of its day – 16 hours 6 minutes
the tilt of its axis – 28.3°

Saturn – duration of its day – 10 hours 34 minutes
the tilt of its axis – 26.73°

2. Assertion (A) : To orbit around the Sun the planet Mars takes lesser time than the time taken by the Earth.

Reason (R) : The diameter of the planet Mars is less than that of Earth.

Code :

- (a) Both (A) and (R) are true, individually true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are individually true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false
(d) (A) is false, but (R) is true

I.A.S. (Pre) 2006

Ans. (d)

The time taken by any planet to orbit around the Sun depends on the distance between the Sun and the planet. The distance between the Sun and Mars is 227943824 km (228 million km), while the Earth is at a distance of 149598262 km (149.6 million km) from the Sun and the orbital period of Earth is 365.256 days and that of Mars is 687.0 days. Thus statement (A) is false. The diameter of Mars is 6792 km while Earth's equatorial diameter is about 12756 km. Thus Reason (R) is correct.

3. Which one of the following conditions is most relevant for the presence of life on Mars?

- (a) Atmospheric composition
(b) Thermal conditions